

1.Reverse a string without slicing:

>> "Hello World"

s="python world"

print(s)

s[::-1]

2.Count words in a sentence:

>> "A Quick Brown Fox Jump Over the Lazy Dog"

s="A quick Brown Fox Jump Over the Lazy Dog"

print(s.count(p))

30

3.Remove duplicate characters.

>> "Carrot"

s "carrot"

result = " "

if ch in data:

result+=(ch)

print(result)

4.What is difference between lists and sets?

set

using brackets{}

Faster

Mutable

Unordered

Unindexed

list

using brackets[]

Slower

Mutable

Ordered

Indexed

5.Can sets contains lists? If Yes How If no why?

no set cannot contain list

Elements of a set must be hashable (their hash value must not change during their creation).

Lists are mutable (you can add, remove, or modify items), so they are not hashable.

Ex: s={1,2[1,3]}

6.Why sets are faster?

Sets is a collection of data type used to store mutiple unique elements

sets are faster because thier membership checks

ex: x in my list()

sets are mutable

7.Difference between remove () and discard ()

remove()

discard()

raise a key error() does not raise a key error

15. How many methods to access value in dictionary?

Using Key[]

Using .get()

Using Key and-Value()

key.get()

1

16. How many methods to delete elements in dictionary?

pop(n)

popitem()

del()

clear()

1

17. How to perform mathematical operations in sets write method name?

Union(|)

intersection(&)

difference(-)

symmetric difference(^)

1

18. What is packing variable in tuple? Explain using example?

A packing variable in tuple means more and mix value store in one single variable

t=(1,"john",25.5,"True")

1

19. What is unpacking variables in tuple? Explain using example?

Tuple unpacking (also called destructuring)

name,age,job = data

pprint(name)

print(age)

print(job)

1

20. Difference between sets and dictionary?

sets

using{}

mutable

unordered

dictionary

using{}

mutable

key-value pair

1

comparision

```
print(10 == 10) # True
```

```
print(10 != 5) # True
```

```
print(10 > 5) # True
```

```
print(10 < 5) # False
```

Assignment

```
x = 5
```

```
x = 2
```

membership

```
(30 in nums)
```

```
(30 in not nums)
```

11. Write precedence Rule of arithmetic in Python?

Take input from user.

Calculate arithmetic operation with operand

Print the result

Higher
Lower X

12. How to sort elements in list using slicing method?

```
>> [5, 11, 14, 7, 8, 3, 2, 5, 9]
```

```
lst = [5, 11, 14, 7, 8, 3, 2, 5, 9]
```

```
sorted_lst = sorted(lst[:]) # slice creates a copy  
print(sorted_lst)
```

lst.sort() X

13. What is Dictionary

A Dictionary in python is built-in data-type used to store data in the form of key-value pair

✓ (1)

14. How many methods to create dictionary?

A Dictionary in python is built-in data-type used to store data in the form of key-value pair

Using Two Methods

1. {} - curly braces

2. dict() - constructor

3. Using Dict() with tuple keys

4. Dict Comprehension

(1)

```
s={1,2,3}
print(s)
s.remove(3)
s.remove(3)
#gets a key error
```

```
discard
s={1,2,3 4,5}
print(s)
s.discard(4)
s.discard(4)
does not raise a key error
```

8.How many methods to insert elements in the list? Total Method?

3 methods

append()-Add one from last element

insert()-insert at specific Index

extend()-Adds multiple elements

9.How many methods to delete elements in the lists? Total Method?

5 methods

remove(): remove by value()

pop(n):remove by index value

pop()

clear()

del()

10.How many Operators in Python Write their name in Explain?

Arithmetic operator: + - % * /

Comparison operator: < , > , ! , = , >= , <=

Logical operator: And, Or , Not

Bitwise Operator: &

Membership operator : In, Not In

Assignment operator: =

Ex: Arithmetic operator

num = a

num = b

print(a+b)

print(a-b)

print(a*b)

print(a/b)

print(a%b)